Static routing IPv4

Agenda

- What is routing
- Static vs Dynamic routing
- IPv4 address structure
- Unicast routing
- Default gateway
- Multiple gateways
- Overlapping networks
- Exercise

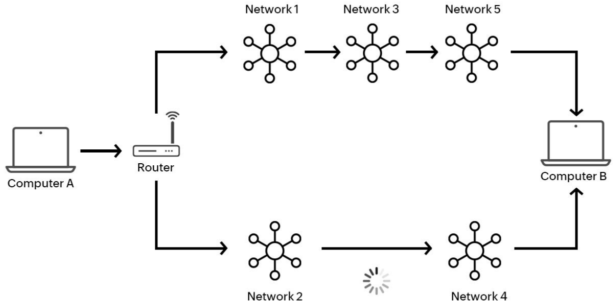
What is routing

Routing is the process of creating a route for data packets.

In computer networks consist of two major components:

- Nodes: these are the machines being used within the network

- Paths or links: these connect nodes to each other



Static vs Dynamic routing

Static routing

Manual entries to routing tables

Dynamic routing

The routing tables are created and updated by routers at runtime based on existing network conditions

Static vs Dynamic routing

Routing Protocol

- In static routing, we don’t use any routing protocol

- Dynamic routing uses distance vector protocols such as RIP and IGRP, and link state protocols such as OSPF and IS-IS to adjust routes

Static vs Dynamic routing

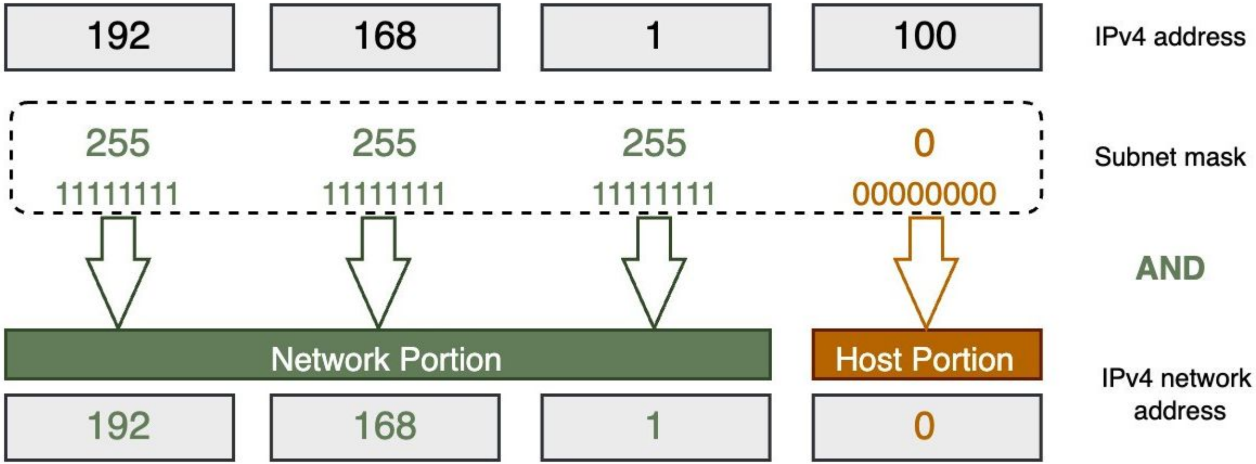
Link failure

- When a failure occurs in a network that uses static routing, this failure affects the whole network

- When a failure occurs in a network that uses dynamic routing, the failure does not affect the whole network

IPv4 address structure

- 32 bit long
- Networks bits
- Host bits



IPv4 address structure

IP address: 10.10.0.**1**/24

Subnet mask: /24 which is 255.255.255.0

10.10.0.0 - Network address

10.10.0.1. - 10.10.0.254 - Hosts part

10.10.0.255 - Broadcast address

IPv4 address structure

IP address: 10.10.**1.1**/16

Subnet mask: /16 which is 255.255.0.0

10.10.0.0 - Network address

10.10.1.1. - 10.10.255.254 - Hosts part

10.10.255.255 - Broadcast address

IPv4 address structure

IP address: 172.16.129.1/18

Subnet mask: /18 which is 255.255.192.0

172.16.128.0 - Network address

172.16.128.1 - 172.16.191.254 - Hosts part

172.16.191.255 - Broadcast address

IPv4 address structure

Address: 172.16.129.1 10101100.00010000.10 000001.00000001

Netmask: 255.255.192.0 = 18 11111111.11111111.11 000000.00000000

=>

Network: 172.16.128.0/18 10101100.00010000.10 000000.00000000

Broadcast: 172.16.191.255 10101100.00010000.10 111111.11111111

HostMin: 172.16.128.1 10101100.00010000.10 000000.00000001

HostMax: 172.16.191.254 10101100.00010000.10 111111.11111110

Unicast routing

1. Before sending the packet, device calculates if the destination hosts belongs to different network

2. Source Device Sends Packet

2. Router Receives Packet

3. Routing Table Lookup

4. Forwarding Decision

5. Packet Travels Hop by Hop

6. Destination Receives Packet

Unicast routing

- Source: **192.168.1.12/24**
- Destination: **192.168.1.253/24**

Unicast routing

- Source: **192.168.1.12/24**
- Destination: **192.168.2.253/24**

Unicast routing

- Source: **192.168.1.12/20**
- Destination: **192.168.10.253/20**

Unicast routing

- Source: **192.168.1.12/20**
  - **11000000.10101000.0000**0001.00001100
- Destination: **192.168.10.253/20**
  - **11000000.10101000.0000**1010.00001100

Default gateway & default route

A **default gateway** is the node that serves as the forwarding host (router) to other networks when no other route specification matches the destination IP address of a packet.

Default route is a routing table entry that matches all destinations not found in the routing table (typically written as 0.0.0.0/0 for IPv4).

```
ip route

default via 172.20.0.1 dev eth0
```

Routing table

```
root@9810cb750610:/# route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         172.20.0.1      0.0.0.0         UG    0      0        0 eth0
172.20.0.0      0.0.0.0         255.255.0.0     U     0      0        0 eth0
172.21.0.0      0.0.0.0         255.255.0.0     U     0      0        0 eth1
172.22.0.0      172.21.0.4      255.255.0.0     UG    0      0        0 eth1
```

```
root@9810cb750610:/# ip route
default via 172.20.0.1 dev eth0
172.20.0.0/16 dev eth0 proto kernel scope link src 172.20.0.3
172.21.0.0/16 dev eth1 proto kernel scope link src 172.21.0.3
172.22.0.0/16 via 172.21.0.4 dev eth1
```

Multiple gateways

Host can have more than one gateway depending on the destination

```
root@ubuntu22:~# ip route add 6.6.6.0/24 via 192.168.100.200 dev eth0
root@ubuntu22:~# ip r
default via 192.168.100.1 dev eth0 proto static
6.6.6.0/24 via 192.168.100.200 dev eth0
```

Multiple gateways

What if we have more than one gateway to one destination?

Multiple gateways

What if we have more than one gateway to one destination?

```
6.6.6.0/24 via 192.168.100.200 dev eth0
6.6.6.0/24 via 192.168.100.201 dev eth0 metric 20
6.6.6.0/24 via 192.168.100.202 dev eth0 metric 30
```

Overlapping networks

Let’s say we have two routes in routing table:

ip route add 10.10.1.0/24 via GATEWAY_A

ip route add 10.10.0.0/16 via GATEWAY_B

ip route add default via GATEWAY_C

And we want to ping: **10.10.1.1**

10.10.1.1 matches all routes:

It’s in 10.10.1.0/24 (covers 10.10.1.0–10.10.1.255)

It’s also in 10.10.0.0/16 (covers 10.10.0.0–10.10.255.255)

It’s also in 0.0.0.0/0

Overlapping networks

Let’s say we have two routes in routing table:

ip route add 10.10.1.0/24 via GATEWAY_A metric 40

ip route add 10.10.0.0/16 via GATEWAY_B metric 30

ip route add default via GATEWAY_C metric 10

And we want to ping: **10.10.1.1**

The routing will happen?....

Overlapping networks

Let's say we have two routes in routing table:

ip route add 10.10.1.0/24 via GATEWAY_A metric 40

ip route add 10.10.0.0/16 via GATEWAY_B metric 30

ip route add default via GATEWAY_C metric 10

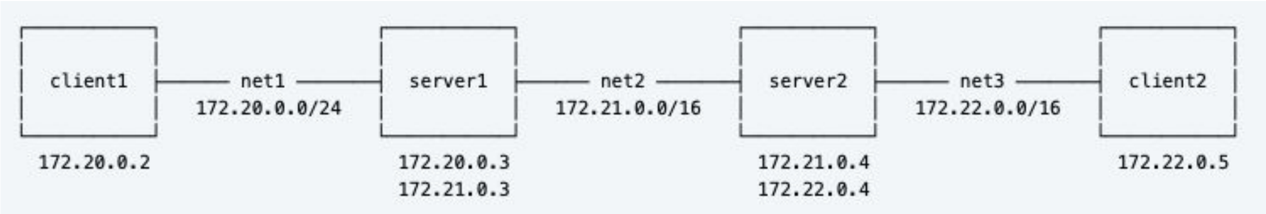
And we want to ping: **10.10.1.1**

The routing will happen?....

The metric takes into account when we have 1:1 match for the target networks

Exercice

Let’s connect client1 with client2 via routing tables



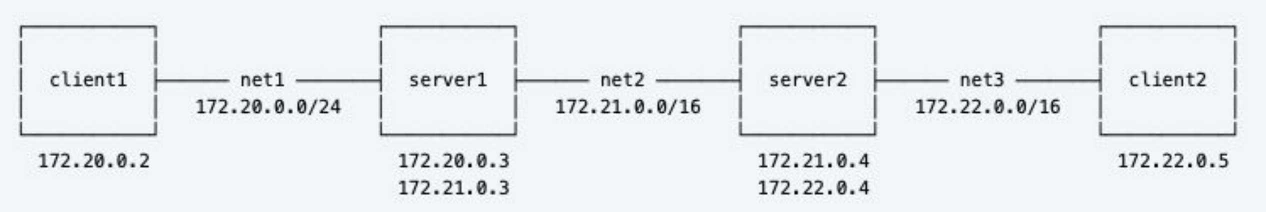
Reference:

ip route add <destination_network>/<prefix> via <next_hop_ip> *metric <number>*

ip route add 10.10.1.0/24 via 192.168.1.1 *metric 20*

Home exercice 1

- Add black hole route on server2 for net3
- Test from client1
- Use tcpdump when necessary
- How is black hole routing different from a firewall DROP rule?
- Why didn’t client1 get an error or “destination unreachable”?
- How would this behavior change if you used a REJECT rule in the firewall instead?



Home exercice 2

- Add server3 container which belongs to the net1 and net2
- Set 2 IP routes with server1 and server3 gateways but same net2 but different metrics
- Test out how it works
- Make server1 unreachable (with lower metric)
- Will it go through ip route with higher metric?